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B.C.A. (Fifth Semester) Examination, 2025

First Paper

(Introduction to DBMS)

Time : Two Hours] [Maximum Marks : 75

Note : Attempt all sections as directed.

Section-A

(Very Short Answer Type Questions)

Note : Attempt all five questions. Each question carries two marks and the answer of each question should not exceed 50 words. $5 \times 2 = 10$

1. Demonstrate transitive dependency.
2. Disadvantage in File Processing System.
3. What is Data Independence?
4. What is the difference between the strong entity set and weak entity set?
5. What do you mean by Integrity constraints?

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Section-B

(Short Answer Type Questions)

Note : Attempt any five questions out of total eight questions. Each question carries five marks and the of each question should not exceed 100 words. $5 \times 5 = 25$

1. Determine the closer of the following set of functional dependencies for a relational scheme $R(A, B, C, D)$ and FDs $\{AB \rightarrow C, C \rightarrow D, D \rightarrow A\}$. List out the candidate keys of R.
2. What is transaction? Explain the ACID properties of transactions.
3. Define Data Manipulation Language (DML) of SQL. List and explain various DML commands.
4. What is Entity-Relationship model? Give symbol used in E-R Diagram.
5. Explain candidate key and primary key with the help of example

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6. What is deadlock? When it occurs?
7. What are the responsibilities of a DBA?
8. What are anomalies in database design?
How can we solve it?

Section-C

(Long Answer Type Questions)

Note : Attempt any **two** question out of **four** questions. Each question carries twenty marks and the answer of each question should not exceed 400 words.

$2 \times 20 = 40$

1. What is normalization? What are the conditions are required for a relation to be in 2NF and 3NF explain with examples?
2. Consider the following tables: Employee
(Emp_no, Name, Emp_city) Company
(Emp_no, Company_name, Salary)
(i) Write a SQL query to display Employee name and company name.

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- (ii) Write a SQL query to display employee name, employee city, company name and salary of all the employees whose salary > 10000
- (iii) Write a query to display all the employees working in 'XYZ' company.
3. Why is concurrency control needed? Explain lost update, Inconsistent retrievals and Un-committed dependency anomalies.
4. Explain ANSI-SPARC 3 Level architecture of DBMS and type of data independence in between different levels. Give the suitable diagram in support of your explanation.

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